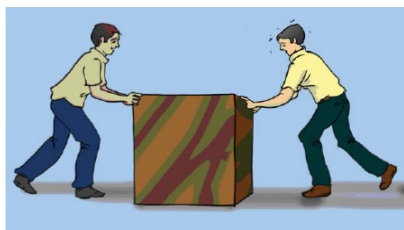




FRICTION -03

Class 08 - Science

1. What is avoided between two surfaces to make movement smooth? [1]
2. What are fluids? [1]
3. Name some objects which have streamlined body. [1]
4. What happens, if the floor we walk on is frictionless? [1]
5. What is the function of ridges on your palm and bottom of your feet? [1]
6. Does the use of ball bearing reduce or increase friction. [1]
7. What is fluid friction? [1]
8. Name the force responsible for wearing out of bicycle tyres. [1]
9. What is used to reduce friction in machines where lubrication are not advisable? [1]
10. Can we eliminate friction completely? [1]
11. Name any machine in which ball bearing is used to reduce friction. [1]
12. What does frictional force exerted on an object in a fluid depend on? [1]
13. Explain why sportsman use shoes with spikes. [1]
14. Why do pieces of luggage fitted with rollers? [1]
15. When does rolling friction come into play? [1]
16. Name the special shape given to objects moving in fluid. [1]
17. What is the cause of friction? [1]
18. How sliding friction is replaced by rolling in machines? [1]
19. Name the substance which is used in carrom board to reduce friction. [1]
20. What is used in cycle brakes to increase friction? [1]
21. On which kind of surface the friction will be greater [1]
 - (i) smooth surface.
 - (ii) rough surface.
22. What is drag? [1]
23. What do we use in cycle brakes to increase friction? [1]
24. Enlist any four methods of reducing friction. [2]
25. Suppose your writing desk is tilted a little. A book kept on it starts sliding down. Show the direction of frictional force acting on it. [2]
26. Describe some examples where heat is produced due to friction. [2]
27. Fig. shows two boys applying force on a box. If the magnitude of the force applied by each is equal, will the box experience any force of friction? [2]



28. Amit is pushing a heavy trunk on the floor. Finding it a little difficult, he turns the trunk to place it vertically. Will he need to do less effort now? Explain. [2]
29. Will the force of friction come into play when a raindrop rolls down a glass window plane? [2]
30. Two blocks of iron of different masses are kept on a cemented floor as shown in Fig. Which one of them would require a larger force to move it from the rest position? [2]



31. Describe some events when we have reduce the friction. [2]
32. Name two factors on which the magnitude of the frictional force depends? [2]
33. Name the two factors on which the magnitude of the frictional force depends. [2]
34. Explain why objects moving in fluids must have special shapes. [2]
35. You might have noticed that when used for a long time, slippers with rubber soles become slippery. Explain the reason. [2]
36. You spill a bucket of soapy water on a marble floor accidentally. Would it make it easier or more difficult for you to walk on the floor? Why? [2]
37. Describe some examples where we deliberately increase friction. [2]
38. What is a spring balance? [2]
39. List the name of five smooth surfaces and five rough surfaces from your surroundings. [2]
40. Why do we shape aeroplanes like that of a bird? [2]
41. Explain why it is convenient to pull the luggages fitted with rollers. [2]
42. You would not have forgotten the accident that happened to astronaut Kalpana Chawla. What was the reason behind that? [2]
43. Iqbal has to push a lighter box and Seema has to push a similar heavier box on the same floor. Who will have to apply a larger force and why? [2]
44. Describe some events where friction is harmful to us. [2]
45. On what factors does the fluid friction depend on? [2]
46. Why objects given special shape which is moving in fluids? [2]
47. Why the soles of our shoes and tyres of cars, trucks etc are grooved? [2]
48. Explain why we use ball bearings between the hub and axle of ceiling fan and bicycle. [2]
49. A marble is allowed to roll down an inclined plane from a fixed height. At the foot of the inclined plane, it moves on a horizontal surface [2]
- covered with silk cloth
 - covered with a layer of sand and
 - covered with a glass sheet.
- On which surface will the marble move the shortest distance. Give a reason for your answer.
50. Explain why sliding friction is less than static friction. [2]
51. On what factors does the fluid friction depends? [2]

52. How the fluid friction can be minimized? [2]
53. While playing tug of war, Preeti felt that the rope was slipping through her hands. Suggest a way out for her to prevent this. [3]



54. What are the factors on which frictional force depends in fluids? [3]
55. What is static and sliding friction? [3]
56. Give examples to show that friction is both a friend and a foe. [3]
57. Is there a force of friction between the wheels of a moving train and iron rails? If yes, name the type of friction. [3]
If an air cushion can be introduced between the wheel and the rail, what effect will it have on the friction?
58. Describe some events where friction is harmful to us? [3]
59. Explain why the surface of mortar and pestle (silbatta) used for grinding is etched again after prolonged use? [3]
60. The handle of a cricket bat or a badminton racquet is usually rough. Explain the reason. [3]
61. We have two identical metal sheets. One of them is rubbed with sandpaper and the other with ordinary paper. [3]
The one rubbed with sandpaper shines more than the other. Give reason.
62. Why can't we write with a chalk if there were no friction? [3]
63. Explain in details why friction is called necessary evil. [3]
64. While travelling on a rickshaw, you might have experienced that if the seat cover is very smooth, you tend to slip [3]
when brakes are applied suddenly. Explain.
65. The soles of our shoes get worn out after a period of time. Explain why? [3]
66. State advantages and disadvantages of fluid friction? [4]
67. The tyres of automobiles are made corrugated and rough. Why? [4]
68. Explain why rolling friction less than sliding friction? [4]
69. a. When you play a game of carom, what do you do to make the coins move smoothly on the board? [4]
b. If you try to hold a glass with soapy hands, it can easily slip from your hands. Why?
70. Show with the help of activity that friction depends on the nature of two surfaces? [4]
71. Show with help of activity that weight of the object increase friction? [4]
72. Ram has to push a lighter box and Shikha has to push a similar heavier box on the floor. Who will have to apply [4]
larger force and why?
73. Explain briefly any two methods of reducing friction? [4]
74. Describe some examples where we deliberately increase friction? [4]
75. Why do kabaddi players rub hands with soil? [4]